BOWEN ZUO

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EDUCATION

University of California, Riverside Sept 2024 – Present
Electrical & Computer Engineering, PhD Student

Georgia Institute of Technology, Atlanta Aug 2022 – May 2024
M.S. in Computer Science
Cumulative GPA: 3.86 / Major GPA: 3.91

University of Washington, Seattle Sept 2018 – June 2022
Electrical & Computer Engineering, Embedded Systems
Cumulative GPA: 3.76 / Major GPA: 3.80

RESEARCH EXPERIENCE

Zhu Lab, UC Riverside, Riverside Sept 2024 – Present

- Conduct research on large language models, spanning test-time scaling, diffusion language models, AI efficiency, and post-training.
- Developed adaptive test-time compute allocation methods for LLMs, including bandit-based budget allocation and evolving in-context demonstrations (ICLR 2026, Findings of ACL 2026).
- Designed test-time scaling and RL post-training methods for diffusion language models, including frequency-aware remasking (NeurIPS 2026) and risk-sensitive GRPO (COLM Non-AR Workshop).

AVA Lab, Georgia Institute of Technology, Atlanta Jan 2023 – May 2024

- Led work on talking-face generation and surveyed recent digital-human research and applications.
- Investigated audio-to-gesture modeling, with a focus on dyadic conversation settings.

ACME Lab, University of Washington, Seattle Oct 2021 – June 2022

- Worked on High-Level Synthesis for Machine Learning (hls4ml) and built an ML algorithm accelerator.

PROJECTS

Test-time Scaling with LLMs

- Scaled LLM inference for math reasoning and code generation by developing a dynamic compute-allocation algorithm that replaced uniform inference budgets, which achieved consistent performance gains across evaluated benchmarks. Extended the base method with bandit-based adaptive allocation, improving robustness on skewed, hard-question-dominant distributions; this work became an ICLR 2026 accepted paper.
- Developed an adaptive allocation framework that jointly decides where to spend test-time compute and how to generate, conditioning on successful responses to semantically related queries as evolving in-context demonstrations; this work was accepted to Findings of ACL 2026.

Diffusion Language Models

- Proposed a two-stage test-time scaling method for diffusion language models: a fast base sampler handles easy questions, and frequency-aware remasking, which tracks how often token predictions change across denoising steps, targets the remaining hard ones. On LLaDA-8B-Instruct and Dream-7B-Instruct, achieved up to 7.4% accuracy gains on MATH-500 and up to 11.18% coverage gains on HumanEval; this work was accepted to NeurIPS 2026.

Post-training

- Developed stepwise risk-sensitive GRPO for diffusion language models to encourage exploration and improve reasoning during RL post-training; this work appeared at the COLM Non-AR Workshop 2026.

· Analyzed why on-policy distillation fails to transfer capabilities between language models, identifying vanishing learning signals that cause early loss plateaus with larger-scale teachers on code generation and math reasoning.

3D Talking Face Generative Models

· Reproduced training of a 3D talking-face model on Meta's Multiface dataset using PyTorch3D.
· Added Multiface facial textures to models that originally produced texture-free outputs.
· Built an interactive pipeline using Google TTS and the ChatGPT API to generate talking-face video responses.
· Worked with multimodal embeddings (identity, audio, images, and text) in encoder–decoder architectures.

TEACHING EXPERIENCE

Teaching Assistant, CS 4644/7643 Deep Learning, Georgia Institute of Technology.

Head Teaching Assistant, CS 228 Introduction to Deep Learning, UC Riverside.

Teaching Assistant, CS 260 Large Models and Advances in AI, UC Riverside.

PUBLICATIONS

B. Zuo, Y. Yu, D. Zhou, Y. Zhu. "Test-time Scaling for Diffusion Language Models with Frequency-Aware Remasking." NeurIPS 2026.

B. Zuo, Y. Zhu. "Strategic Scaling of Test-Time Compute: A Bandit Learning Approach." ICLR 2026.

B. Zuo, D. Zhou, Y. Zhu. "Adaptive Test-Time Compute Allocation with Evolving In-Context Demonstrations." Findings of ACL 2026.

Y. Yu, **B. Zuo**, D. J. Crandall, Y. Zhu, D. Zhou. "Exploring More, Reasoning Better: Stepwise Risk-Sensitive GRPO for Diffusion Language Models." COLM Non-AR Workshop 2026.

L. Zhao, Q. Zhao, **B. Zuo**, Q. Zhan. "Why On-Policy Distillation Sometimes Fails: Vanishing Learning Signals." arXiv:2610.11247, 2026.

F. Wu, D. Xing, Y. Huang, et al. (incl. **B. Zuo**). "UniEvo-VL: An On-policy Self-Distillation Training Recipe for Multimodal Model Self-improvement." arXiv:2609.38721, 2026.

Coauthor. "Conversational Gestures: Transforming Text into Full-Body Virtual Interactions." SouthNLP, 2024.

Z. Jiang, B. Carlson, **B. Zuo**, et al. "Machine Learning Evaluation in the Global Event Processor FPGA for the ATLAS Trigger Upgrade." Journal of Instrumentation (JINST), 2024.